

June 22, 2026

Junior Research Group in AI in Biomedical Imaging
Luxembourg Centre for Systems Biomedicine (LCSB)
University of Luxembourg

Dear Members of the Selection Committee,

I am writing to express my strong interest in the Doctoral Researcher position focusing on AI and Machine Learning for Biomedical Imaging. As an AI & Data Science Engineer with extensive academic experience in translating complex medical data into predictive deep learning architectures, I am highly motivated to contribute to your research on neurodegenerative diseases.

Throughout my engineering theses, I have focused heavily on building robust and trustworthy AI models for clinical applications. For my Master's thesis on the "Cardiovascular Medical Digital Twin," I worked extensively with complex medical parameters and multimodal inputs to predict critical health outcomes. I engineered Medical_MambaTS, a linear-time state-space model that processes high-dimensional, irregular clinical data without the quadratic bottleneck of standard transformers. More importantly, I developed PI-DeepONet, a Physics-Informed Operator Network that maps the strict mathematical constraints of hemodynamics directly into the neural architecture.

This experience of embedding structural deterministic rules into "black-box" neural networks via PyTorch Autograd is directly applicable to the challenges of biomedical image computing. When developing state-of-the-art methods for image segmentation and detection—especially for complex neurodegenerative tracking—ensuring that the AI model is trustworthy and robust against data sparsity is paramount. I am highly proficient in Python, PyTorch, and navigating complex High-Performance Computing (HPC) environments for model training.

I am eager to transition my background in medical digital twins and continuous-time modeling toward computer vision and pattern recognition at the LCSB. Please find attached my CV and academic transcripts for your review.

I look forward to the possibility of discussing how my skills align with your research goals.

Sincerely,
Ismail Aissaoui
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